From: prb@aps.org
Subject: Your_manuscript BN15047 Saito
Date: March 19, 2026 at 3:18
To: hiroto.saito.t5@dc.tohoku.ac.jp

Re: BN15047
First-principles analysis of in-plane anomalous Hall effect using symmetry-adapted Wannier Hamiltonians and multipole decomposition
by Hiroto Saito and Takashi Koretsune

Dear Dr. Saito,

Thank you for submitting your manuscript to Physical Review B. We appreciate your interest and the opportunity to consider your work.

We have completed this round of review of your manuscript, and the reviewer comments are attached. They have raised substantial issues. We ask that you address these concerns through major revisions and provide a clear, detailed response so that we may proceed with further consideration.

Please use https://authors.aps.org/Submissions/ if you resubmit. Be sure to include a complete response to the reviewer comments, a marked-up PDF of the manuscript showing changes, and a list of those changes. We only need the PDF file of the manuscript, not the TeX or Word source file or figure files.

Yours sincerely,

Dr. Yonko Millev
Editor
Physical Review B
Email: prb@aps.org
https://journals.aps.org/prb/

NEWS FROM THE PHYSICAL REVIEW JOURNALS

We now request titles in the reference list of papers
https://go.aps.org/prbtitles

PR and PRL encourage "joint" submissions
https://go.aps.org/3MzNENg

P.S. We regret the delay in obtaining these reports.

P.S. Another referee we consulted was not able to review your manuscript.

PROBLEMS WITH MANUSCRIPT:

The following problems were noted in your manuscript.

* Please add a numbered reference corresponding to the data mentioned in the data availability statement to the reference list of the manuscript. Please also update the data availability statement to cite the reference(s) by providing the reference number(s) when prompted. Thank you for your support of our data sharing initiative. More information can be found at https://go.aps.org/das.

* Thank you for participating in our data sharing initiative. If you are unable or unwilling to provide the data publicly for any reason, one simple solution would be to change the statement to the one ending "... data available from the authors upon reasonable request." More information can be found at https://go.aps.org/das.

Report of the First Referee -- BN15047/Saito

The authors present a symmetry-resolved multipole decomposition of a first-principles Wannier Hamiltonian and applies this framework to analyze the in-plane anomalous Hall effect (IAHE) in bcc Fe. By constructing a symmetry-adapted multipole basis (SAMB), the authors decompose the Hamiltonian into electric (Q), magnetic (M), magnetic toroidal (T), and electric toroidal (G) multipole components. They then analyze the angular dependence of the anomalous Hall conductivity by selectively rotating multipole channels of different ranks. The work argues that higher-rank magnetic and magnetic toroidal multipoles play a quantitatively significant role in shaping the angular structure of IAHE, and that strain can enhance specific symmetry channels.

The manuscript provides a systematic symmetry-based decomposition framework and presents detailed numerical results. However, several issues should be clarified.

1. The manuscript is currently overly detailed in sections that are not central to the new physical message. 1) Equations (1)-(15), including the formal derivation of IAHE, appear unnecessarily extended. Since the anomalous Hall conductivity formula is standard, it would be sufficient to state the final expression and summarize the symmetry arguments. The detailed derivation could be moved to the Supplementary Material. 2) The time-reversal symmetric Wannier construction and symmetry-adapted Wannier methodology have already been discussed in previous publications. A concise summary with references would improve readability.

2. The central results rely on rotating the magnetization direction and analyzing the angular dependence of IAHE. However, the treatment of the magnetic ground state during this rotation is not sufficiently clarified.

3. The anomalous Hall conductivity is ultimately computed from the Berry curvature obtained from the full Hamiltonian. The multipole decomposition does not modify the Hall conductivity formula, nor does it alter the numerical prediction of σ. Rather, it reorganizes the Hamiltonian into symmetry-classified channels and attributes different angular harmonics to different multipole ranks. While such a decomposition may provide a structured interpretation of angular dependence, it is not clear whether it leads to new experimentally testable predictions beyond what can already be understood from conventional analyses in terms of spin–orbit coupling and band anisotropy.

Report of the Second Referee -- BN15047/Saito

This manuscript presents a new method for studying and analyzing the anomalous Hall effect in magnetic systems. The model is constructed using a tight-binding Hamiltonian derived from Wannier functions projected onto a symmetry-adapted multipole basis (SAMB).

A further improvement in the fidelity of the model comes from the usage of Wannier functions, which are time-reversal symmetric. The authors show how the model can be split into an electric, time-reversal-symmetric part and a magnetic, time-reversal-antisymmetric part, and how the latter can be rotated to model different magnetisation directions. For bcc iron, the method is demonstrated to be effective at reproducing the magnetisation-direction dependence in good agreement with direct DFT calculations for different orientations.

The presented method is sound and of high conceptual importance, resonating with current efforts in the literature to extend symmetry analysis to magnetic and topological systems.

Although I greatly appreciate the authors' methodological and theoretical work, I have found the paper very hard to read and, in many respects, ineffective in presenting the method and theory. The content of the manuscript relies too strongly on the content of previous papers.

* In section II-A, the distinction between general formulas and those applying to the Iron-BCC should be made clearer.

* Section II-B relies strongly on ref 37, even if I admit it can be difficult to introduce briefly such a technical part, the readability of the paper would benefit greatly from a more gradual introduction of the SAMB, also no technical detail on how the SAMB projection of the Wannier Hamiltonian is implemented. Very few technical details are discussed in this paper, and all are referred to previous works.

The results and discussion are a bit too scattered and erratic to make an effective impact on the reader.

I recommend the publication of the paper after a revision of the content and the inclusion of more technical details